

Step 1 : Import Required Libraries

```
!pip install folium
```

[Show hidden output](#)

```
import pandas as pd
import numpy as np
import folium
```

Step 2 : Load the Dataset

```
df = pd.read_csv('/content/Dataset .csv')
```

df

|      | Restaurant ID | Restaurant Name          | Country Code | City             | Address                                           | Locality                                   | Locality Verbose                                  | Longitude  | Latitude  | Cuisines                         | ... | Currency         | Has Table booking | Has Online delivery | Is delivering now | Switch to order menu | Price range | Aggregate rating | Rating color |
|------|---------------|--------------------------|--------------|------------------|---------------------------------------------------|--------------------------------------------|---------------------------------------------------|------------|-----------|----------------------------------|-----|------------------|-------------------|---------------------|-------------------|----------------------|-------------|------------------|--------------|
| 0    | 6317637       | Le Petit Souffle         | 162          | Makati City      | Third Floor, Century City Mall, Kalayaan Avenu... | Century City Mall, Poblacion, Makati City  | Century City Mall, Poblacion, Makati City, Mak... | 121.027535 | 14.565443 | French, Japanese, Desserts       | ... | Botswana Pula(P) | Yes               | No                  | No                | No                   | 3           | 4.8              | G            |
| 1    | 6304287       | Izakaya Kikufuji         | 162          | Makati City      | Little Tokyo, 2277 Chino Rocas Avenue, Legaspi... | Little Tokyo, Legaspi Village, Makati City | Little Tokyo, Legaspi Village, Makati City, Ma... | 121.014101 | 14.553708 | Japanese                         | ... | Botswana Pula(P) | Yes               | No                  | No                | No                   | 3           | 4.5              | G            |
| 2    | 6300002       | Heat - Edsa Shangri-La   | 162          | Mandaluyong City | Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal... | Edsa Shangri-La, Ortigas, Mandaluyong City | Edsa Shangri-La, Ortigas, Mandaluyong City, Ma... | 121.056831 | 14.581404 | Seafood, Asian, Filipino, Indian | ... | Botswana Pula(P) | Yes               | No                  | No                | No                   | 4           | 4.4              | G            |
| 3    | 6318506       | Ooma                     | 162          | Mandaluyong City | Third Floor, Mega Fashion Hall, SM Megamall, O... | SM Megamall, Ortigas, Mandaluyong City     | SM Megamall, Ortigas, Mandaluyong City, Mandal... | 121.056475 | 14.585318 | Japanese, Sushi                  | ... | Botswana Pula(P) | No                | No                  | No                | No                   | 4           | 4.9              | G            |
| 4    | 6314302       | Sambo Kojin              | 162          | Mandaluyong City | Third Floor, Mega Atrium, SM Megamall, Ortigas... | SM Megamall, Ortigas, Mandaluyong City     | SM Megamall, Ortigas, Mandaluyong City, Mandal... | 121.057508 | 14.584450 | Japanese, Korean                 | ... | Botswana Pula(P) | Yes               | No                  | No                | No                   | 4           | 4.8              | G            |
| ...  | ...           | ...                      | ...          | ...              | ...                                               | ...                                        | ...                                               | ...        | ...       | ...                              | ... | ...              | ...               | ...                 | ...               | ...                  | ...         | ...              | ...          |
| 9546 | 5915730       | Nami Gurme               | 208          | İstanbul         | Kemanköy Paşa Mahallesi, R'ht'm ...               | Karaköy                                    | Karaköy, İstanbul                                 | 28.977392  | 41.022793 | Turkish                          | ... | Turkish Lira(TL) | No                | No                  | No                | No                   | 3           | 4.1              | G            |
| 9547 | 5908749       | Ceviz Aca                | 208          | İstanbul         | Koşuyolu Mahallesi, Muhtittin ...                 | Koşuyolu                                   | Koşuyolu, İstanbul                                | 29.041297  | 41.009847 | World Cuisine, Patisserie, Cafe  | ... | Turkish Lira(TL) | No                | No                  | No                | No                   | 3           | 4.2              | G            |
| 9548 | 5915807       | Huqqa                    | 208          | İstanbul         | Kuruçeşme Mahallesi, Muallim Naci Caddesi, N...   | Kuruçeşme                                  | Kuruçeşme, İstanbul                               | 29.034640  | 41.055817 | Italian, World Cuisine           | ... | Turkish Lira(TL) | No                | No                  | No                | No                   | 4           | 3.7              | Y            |
| 9549 | 5916112       | Ak Kahve                 | 208          | İstanbul         | Kuruçeşme Mahallesi, Muallim Naci Caddesi, N...   | Kuruçeşme                                  | Kuruçeşme, İstanbul                               | 29.036019  | 41.057979 | Restaurant Cafe                  | ... | Turkish Lira(TL) | No                | No                  | No                | No                   | 4           | 4.0              | G            |
| 9550 | 5927402       | Walter's Coffee Roastery | 208          | İstanbul         | Cafeağa Mahallesi, Bademaltı Sokak, No 21/B, ...  | Moda                                       | Moda, İstanbul                                    | 29.026016  | 40.984776 | Cafe                             | ... | Turkish Lira(TL) | No                | No                  | No                | No                   | 2           | 4.0              | G            |

9551 rows x 21 columns

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Restaurant ID        9551 non-null   int64
1   Restaurant Name      9551 non-null   object
2   Country Code         9551 non-null   int64
3   City                 9551 non-null   object
4   Address              9551 non-null   object
5   Locality              9551 non-null   object
6   Locality Verbose     9551 non-null   object
7   Longitude             9551 non-null   float64
8   Latitude              9551 non-null   float64
9   Cuisines              9542 non-null   object
10  Average Cost for two  9551 non-null   int64
11  Currency              9551 non-null   object
12  Has Table booking     9551 non-null   object
13  Has Online delivery   9551 non-null   object
14  Is delivering now     9551 non-null   object
15  Switch to order menu  9551 non-null   object
16  Price range           9551 non-null   int64
17  Aggregate rating      9551 non-null   float64
18  Rating color          9551 non-null   object
19  Rating text           9551 non-null   object
20  Votes                 9551 non-null   int64
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

Step 3 : Data Preprocessing

1. Feature selection

```
df_geo = df [[
    "Restaurant ID",
    "Restaurant Name",
    "City",
    "Locality",
    "Locality Verbose",
    "Latitude",
    "Longitude",
    "Cuisines",
    "Aggregate rating",
```

```
"Price range",
"Average Cost for two"
]]
```

df\_geo

|      | Restaurant ID | Restaurant Name          | City             | Locality                                   | Locality Verbose                                  | Latitude  | Longitude  | Cuisines                         | Aggregate rating | Price range | Average Cost for two |
|------|---------------|--------------------------|------------------|--------------------------------------------|---------------------------------------------------|-----------|------------|----------------------------------|------------------|-------------|----------------------|
| 0    | 6317637       | Le Petit Souffle         | Makati City      | Century City Mall, Poblacion, Makati City  | Century City Mall, Poblacion, Makati City, Mak... | 14.565443 | 121.027535 | French, Japanese, Desserts       | 4.8              | 3           | 1100                 |
| 1    | 6304287       | Izakaya Kikufuji         | Makati City      | Little Tokyo, Legaspi Village, Makati City | Little Tokyo, Legaspi Village, Makati City, Ma... | 14.553708 | 121.014101 | Japanese                         | 4.5              | 3           | 1200                 |
| 2    | 6300002       | Heat - Edsa Shangri-La   | Mandaluyong City | Edsa Shangri-La, Ortigas, Mandaluyong City | Edsa Shangri-La, Ortigas, Mandaluyong City, Ma... | 14.581404 | 121.056831 | Seafood, Asian, Filipino, Indian | 4.4              | 4           | 4000                 |
| 3    | 6318506       | Ooma                     | Mandaluyong City | SM Megamall, Ortigas, Mandaluyong City     | SM Megamall, Ortigas, Mandaluyong City, Mandal... | 14.585318 | 121.056475 | Japanese, Sushi                  | 4.9              | 4           | 1500                 |
| 4    | 6314302       | Sambo Kojin              | Mandaluyong City | SM Megamall, Ortigas, Mandaluyong City     | SM Megamall, Ortigas, Mandaluyong City, Mandal... | 14.584450 | 121.057508 | Japanese, Korean                 | 4.8              | 4           | 1500                 |
| ...  | ...           | ...                      | ...              | ...                                        | ...                                               | ...       | ...        | ...                              | ...              | ...         | ...                  |
| 9546 | 5915730       | Namlı Gurme              | İstanbul         | Karaköy                                    | Karaköy, İstanbul                                 | 41.022793 | 28.977392  | Turkish                          | 4.1              | 3           | 80                   |
| 9547 | 5908749       | Ceviz Aca                | İstanbul         | Koşuyolu                                   | Koşuyolu, İstanbul                                | 41.009847 | 29.041297  | World Cuisine, Patisserie, Cafe  | 4.2              | 3           | 105                  |
| 9548 | 5915807       | Huqqa                    | İstanbul         | Kuruçeşme                                  | Kuruçeşme, İstanbul                               | 41.055817 | 29.034640  | Italian, World Cuisine           | 3.7              | 4           | 170                  |
| 9549 | 5916112       | Ak Kahve                 | İstanbul         | Kuruçeşme                                  | Kuruçeşme, İstanbul                               | 41.057979 | 29.036019  | Restaurant Cafe                  | 4.0              | 4           | 120                  |
| 9550 | 5927402       | Walter's Coffee Roastery | İstanbul         | Moda                                       | Moda, İstanbul                                    | 40.984776 | 29.026016  | Cafe                             | 4.0              | 2           | 55                   |

9551 rows × 11 columns

Next steps: [Generate code with df\\_geo](#) [New interactive sheet](#)

df\_geo.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 11 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Restaurant ID       9551 non-null  int64
1   Restaurant Name     9551 non-null  object
2   City                9551 non-null  object
3   Locality            9551 non-null  object
4   Locality Verbose    9551 non-null  object
5   Latitude            9551 non-null  float64
6   Longitude           9551 non-null  float64
7   Cuisines            9542 non-null  object
8   Aggregate rating    9551 non-null  float64
9   Price range        9551 non-null  int64
10  Average Cost for two 9551 non-null  int64
dtypes: float64(3), int64(3), object(5)
memory usage: 820.9+ KB
```

2. Handling Missing values

df\_geo.isnull().sum()

|                      |   |
|----------------------|---|
|                      | 0 |
| Restaurant ID        | 0 |
| Restaurant Name      | 0 |
| City                 | 0 |
| Locality             | 0 |
| Locality Verbose     | 0 |
| Latitude             | 0 |
| Longitude            | 0 |
| Cuisines             | 9 |
| Aggregate rating     | 0 |
| Price range          | 0 |
| Average Cost for two | 0 |

dtype: int64

```
df_geo["Cuisines"] = df_geo["Cuisines"].fillna("Unknown")
```

df\_geo.isnull().sum()

|                      |   |
|----------------------|---|
|                      | 0 |
| Restaurant ID        | 0 |
| Restaurant Name      | 0 |
| City                 | 0 |
| Locality             | 0 |
| Locality Verbose     | 0 |
| Latitude             | 0 |
| Longitude            | 0 |
| Cuisines             | 0 |
| Aggregate rating     | 0 |
| Price range          | 0 |
| Average Cost for two | 0 |

dtype: int64

Step 4 : Interactive Geographical Mapping

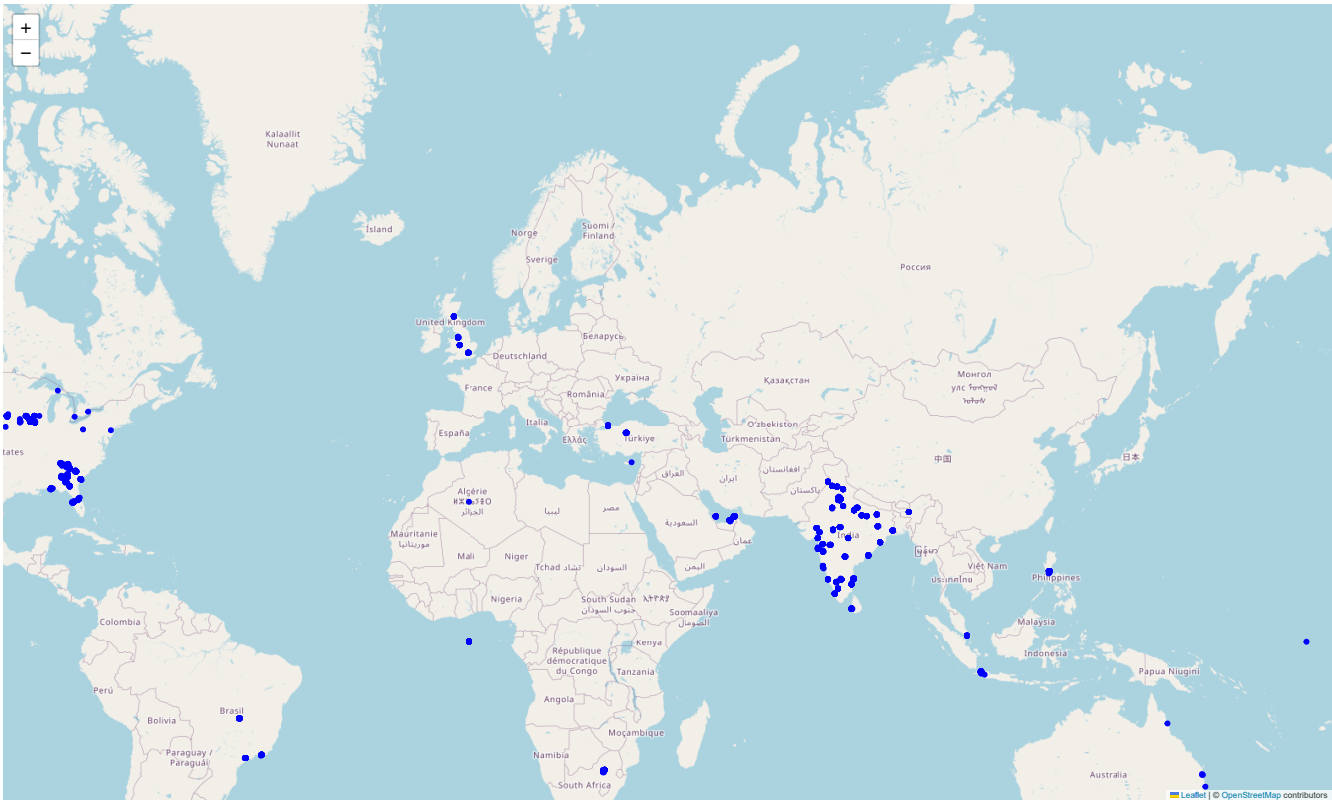
```
map_center = [df_geo["Latitude"].mean(), df_geo["Longitude"].mean()]
```

```
restaurant_map = folium.Map(location=map_center, zoom_start=11)
```

```
for _, row in df_geo.iterrows():
    folium.CircleMarker(
```

```
location=[row["Latitude"], row["Longitude"]],
radius=2,
color="blue",
fill=True,
fill_opacity=0.7,
popup=row["Restaurant Name"] # optional tooltip
).add_to(restaurant_map)
```

restaurant\_map



Step 5 : Group by City & Analyze Concentration

1. Group by City & Analyze Concentration

```
city_counts = df_geo["City"].value_counts()
```

city\_counts

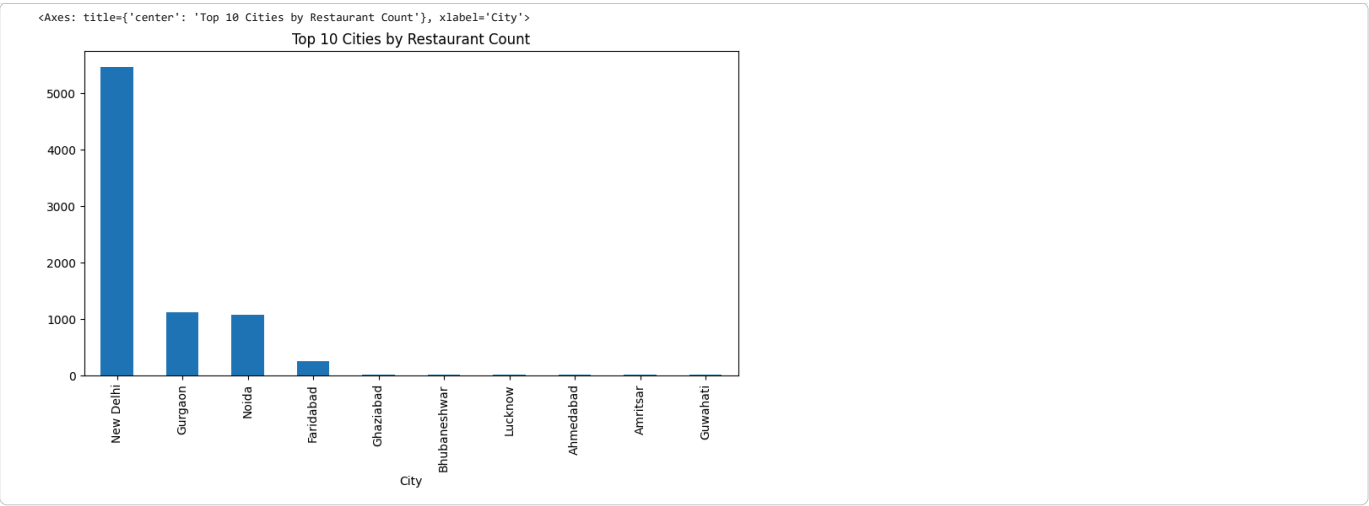
| count          |      |
|----------------|------|
| City           |      |
| New Delhi      | 5473 |
| Gurgaon        | 1118 |
| Noida          | 1080 |
| Faridabad      | 251  |
| Ghaziabad      | 25   |
| ...            | ...  |
| Lakes Entrance | 1    |
| Mohali         | 1    |
| Panchkula      | 1    |
| Bandung        | 1    |
| Randburg       | 1    |

141 rows × 1 columns

dtype: int64

2. Visualize the Top N Cities

```
city_counts.head(10).plot(kind="bar", figsize=(10,5), title="Top 10 Cities by Restaurant Count")
```



This chart shows the top 10 cities with the highest number of restaurants in the dataset. Higher counts indicate areas with strong restaurant presence and potentially higher competition or consumer demand.

```
#Compute percentage share of restaurants per city

city_percentage = (city_counts / city_counts.sum()) * 100
city_percentage.head(10)
```

| City        | count     |
|-------------|-----------|
| New Delhi   | 57.302900 |
| Gurgaon     | 11.705581 |
| Noida       | 11.307716 |
| Faridabad   | 2.627997  |
| Ghaziabad   | 0.261753  |
| Bhubaneswar | 0.219872  |
| Lucknow     | 0.219872  |
| Ahmedabad   | 0.219872  |
| Amritsar    | 0.219872  |
| Guwahati    | 0.219872  |

dtype: float64

Step 6 : Location-Specific Statistics

1. Average Rating per City

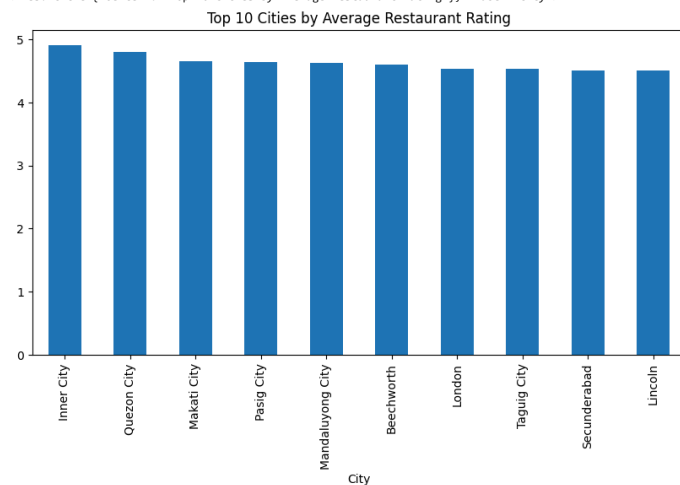
```
avg_rating_by_city = df_geo.groupby("City")["Aggregate rating"].mean().sort_values(ascending=False)
avg_rating_by_city.head(10)
```

| City             | Aggregate rating |
|------------------|------------------|
| Inner City       | 4.900000         |
| Quezon City      | 4.800000         |
| Makati City      | 4.650000         |
| Pasig City       | 4.633333         |
| Mandaluyong City | 4.625000         |
| Beechworth       | 4.600000         |
| London           | 4.535000         |
| Taguig City      | 4.525000         |
| Secunderabad     | 4.500000         |
| Lincoln          | 4.500000         |

dtype: float64

```
#Plot
avg_rating_by_city.head(10).plot(kind="bar", figsize=(10,5), title="Top 10 Cities by Average Restaurant Rating")
```

<Axes: title={'center': 'Top 10 Cities by Average Restaurant Rating'}, xlabel='City'>



Above plot answers the question "Which cities have better-rated restaurants?"

## 2. Average Cost / Price per City

```
avg_cost_by_city = df_geo.groupby("City")["Average Cost for two"].mean().sort_values(ascending=False)
```

```
avg_cost_by_city.head(10)
```

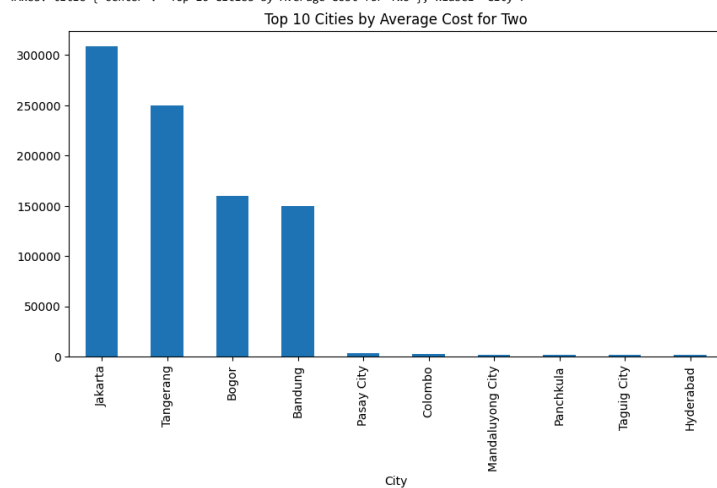
| Average Cost for two |               |
|----------------------|---------------|
| City                 |               |
| Jakarta              | 308437.500000 |
| Tangerang            | 250000.000000 |
| Bogor                | 160000.000000 |
| Bandung              | 150000.000000 |
| Pasay City           | 3333.333333   |
| Colombo              | 2375.000000   |
| Mandaluyong City     | 2000.000000   |
| Panchkula            | 2000.000000   |
| Taguig City          | 1725.000000   |
| Hyderabad            | 1361.111111   |

dtype: float64

```
#Plot
```

```
avg_cost_by_city.head(10).plot(kind="bar", figsize=(10,5), title="Top 10 Cities by Average Cost for Two")
```

<Axes: title={'center': 'Top 10 Cities by Average Cost for Two'}, xlabel='City'>



Above plot answers the question "Which cities are the most expensive for dining?"

## 3. Top Cuisines per City

```
cuisine_city = (
    df_geo.assign(Cuisine=df_geo["Cuisines"].str.split(",")).explode("Cuisine")
)
```

```
top_cuisines_by_city = (
    cuisine_city.groupby(["City", "Cuisine"]).size().reset_index(name="Count")
)
```

```
top_cuisines_by_city
```

|      | City       | Cuisine         | Count |  |
|------|------------|-----------------|-------|--|
| 0    | Abu Dhabi  | Afghani         | 1     |  |
| 1    | Abu Dhabi  | Arabian         | 2     |  |
| 2    | Abu Dhabi  | Asian           | 1     |  |
| 3    | Abu Dhabi  | Biryani         | 1     |  |
| 4    | Abu Dhabi  | Burger          | 1     |  |
| ...  | ...        | ...             | ...   |  |
| 2402 | 🇹🇷istanbul | Desserts        | 2     |  |
| 2403 | 🇹🇷istanbul | Italian         | 1     |  |
| 2404 | 🇹🇷istanbul | Restaurant Cafe | 4     |  |
| 2405 | 🇹🇷istanbul | Turkish         | 1     |  |
| 2406 | 🇹🇷istanbul | World Cuisine   | 1     |  |

2407 rows x 3 columns

Next steps: [Generate code with top\\_cuisines\\_by\\_city](#) [New interactive sheet](#)

```
top_5_cuisines_by_city = (  
    top_cuisines_by_city.sort_values(["City", "Count"], ascending=[True, False]).groupby("City").head(5)  
)  
  
top_5_cuisines_by_city.head(15)
```

|  | City | Cuisine | Count |  |
|--|------|---------|-------|--|
|--|------|---------|-------|--|